Fazle Mohammed Tasfiq

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Profile

I'm a final-year CSE student who likes seeing a project through, not just prototyping it. I've built full-stack platforms (Laravel/Vue, Node/Express) with real payment and coordination workflows, and I'm currently doing ML/NLP research – a multilingual RAG evaluation study and an ongoing medical-image security project across several CNN architectures. Looking for internship or full-time opportunities in software, AI/ML, or research.

Education

BRAC University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; CGPA: 3.78/4.00 *Sept. 2022 – Expected Sept. 2026*

Projects

HelpRelief – Disaster Relief Coordination Platform | *Vue.js, Laravel, PostgreSQL, TailwindCSS* Apr. 2025 – Aug. 2025

- Shipped the volunteer/NGO-staff dashboards – profile management, live campaign and state views, and task activity tracking – for a disaster-relief coordination platform.
- Built the Laravel REST APIs and Eloquent queries behind campaign maps, report aggregation, and task-completion metrics; verified endpoints with Postman during team integration.

Kinder – Babysitter Booking Platform | *React, Node.js, Express, MongoDB, SSLCommerz* Dec. 2025 – Jan. 2026

- Built the SSLCommerz payment flow for completed bookings end to end: payment records, transaction IDs, platform-fee calculation, and gateway callbacks for both cash and online payment.
- Added parent booking-payment screens with success/error/cancel states, wired to protected Express routes and MongoDB payment models.

QA Classification (Course Project) | *Python, Keras, Scikit-learn, NLTK* Apr. 2025 – Aug. 2025

- Implemented a BiLSTM classifier (embedding layer, bidirectional LSTM, dense softmax head) trained on GloVe and Skip-gram embeddings, and benchmarked it against TF-IDF/BoW Logistic Regression, Random Forest, and unidirectional LSTM baselines.

Battleship | *x86 Assembly, EMU8086* May 2025

- Built the majority of a 2-person team's x86 Assembly Battleship game: 8x8 grid with a hidden ship-placement layer, BIOS-clock-seeded randomization, hit/miss state tracking, and a scoring/accuracy system across multiple difficulty levels.

Research

RAG Evaluation for Multilingual QA | *Python, LangChain, LangGraph, LlamaIndex, ChromaDB* July 2025 – Present

- Built the web-scraping, dataset-curation, and LLM-based QA-generation pipeline for the Tamil track of a 5-language (Tamil, Bengali, Hindi, Nepali, Urdu) multilingual RAG benchmark, working across the shared pipeline design with the team.
- Evaluated 4 LLMs on direct question answering, then built and evaluated a RAG pipeline combining ChromaDB retrieval with those LLMs; publication in progress.

Medical Image Security Research | *Python, PyTorch, CNNs, Grad-CAM* Nov. 2025 – Present

- Building an encryption-decryption image-security pipeline across 5 CNN architectures (MobileNetV3, MobileViT, ResNet18, ShuffleNetV2, SqueezeNet) on brain tumor MRI and COVID-19 scans, using chaotic map-based image encryption to stress-test model robustness.
- Generating Grad-CAM interpretability maps and pre/post-attack accuracy comparisons per model as part of an ongoing research paper.

Technical Skills

Languages: Python, Java, PHP, JavaScript/TypeScript, SQL, HTML/CSS, x86 Assembly
Frameworks: Laravel, Vue.js, React, Node.js, Express.js, TailwindCSS
Databases: PostgreSQL, MongoDB, MySQL
ML/NLP: Scikit-learn, Keras, PyTorch, Pandas, NumPy, NLTK, GloVe, Word2Vec, LangChain, LangGraph, LlamaIndex, ChromaDB
Tools: Git, Docker, Postman, Google Colab, Jupyter, VS Code, PyCharm, Cisco Packet Tracer, Arduino